**Semester III** **B. Tech CSE (AI)**

## BCSAI 304: OBJECT ORIENTED PROGRAMMING USING JAVA

**3L + 0T + 0P + 3C** **MM 100**

### Unit: 1

**Introduction to Java**

Introduction, installing java, JRE and JDK, Byte Code, JVM; Simple Java program. Creating Objects, Data types, Operators: Arithmetic Operators, Bitwise operators, Relational operators, Logical Operators, The Assignment Operator, ternary operator; Operator Precedence, Access specifiers. Type casting; Strings. Control Statements: conditional statements, looping statements, jumping statements, methods, static methods, and static block, Arrays

### Unit: 2

**Classes, Inheritance, package and Interface**

Classes: Classes in Java; declaring a class; constructors, method overloading, Object Class.
Inheritance: Simple Inheritance, Super class and sub class, super keyword multiple, and multilevel inheritance; Overriding.
Packages and Interfaces: Packages, Defining Packages, access protection, Importing Packages. Abstract Methods, Abstract Classes, Defining Abstract Classes, Extending Abstract Classes, Defining Interfaces, Implementing Interfaces

**Lambda Expressions**

Introducing Lambda Expressions, Lambda Expression Fundamentals, Functional Interfaces, Some Lambda Expression Examples, Block Lambda Expressions, Generic Functional Interfaces, Passing Lambda Expressions as Arguments

### Unit: 3

**Thread Programming, Exceptions and I/O**

Thread Programming: What are threads? Thread life cycle, Extending Thread class, implementing runnable interface, Synchronization, Deadlock, Manipulation Thread states.
Exception Handling: Fundamentals, Exception Types, try and catch, multiple catch clauses, nested try statements, throw, and throws, finally, built in exceptions, user defined exceptions and chained exceptions. I/O: Introduction, stream classes, Byte Streams, Character Streams, Reading Data from Keyboard, Folders and Folder Operations, Files and File Operations, Serialization and Deserialization

### Unit: 4

**Networking and JDBC**

Networking: Introduction, Socket, Client/Server architecture, Reserved Sockets, Proxy Servers, Internet Addressing, Factory Methods, Instance Methods, TCP/IP Client Sockets : URL,URL Connection, TCP/IP Server Sockets, Datagrams, Datagram Sockets, Inet4Address and Inet6Address, RMI: Stub, Skelton

Database connectivity – JDBC architecture and Drivers. JDBC API - loading a driver, connecting to a database, creating and executing JDBC statements, handling SQL exceptions. Accessing result sets: types and methods. An example - JDBC application to query a database

**Unit: 5**

**GUI Programming with AWT and JavaFX**

AWT: GUI Programming, AWT Basics, AWT package: Layouts, Label, Text Field, Button, Events, Text Area, Checkbox, Checkbox Group, List, Canvas, Menus, Pop Menus, Panel, And Dialog. Listeners: Action Listener, Mouse Listener, Item Listener, Key Listener, Window Listener

JavaFX: Basic Concepts, JavaFX Packages, Stage and Scene Classes ,Nodes and Scene Graphs, Layouts, Application Class and the Lifecycle Methods, Compiling and Running a JavaFX Program, JavaFX Control: Label, Buttons and Events, Event Handling, Button, ToggleButton, RadioButton, Image, ImageView, ListView, ComboBox, Menus, Toolbar, Case Study .

**Text/Reference Books:**


1. Herbert Schildt: Java The Complete Reference, 7th Edition, Tata McGraw Hill, 2007
2. Jim Keogh: J2EE The Complete Reference, Tata McGraw Hill, 2007
3. Y. Daniel Liang: Introduction to JAVA Programming, 6th Edition, Pearson Education, 2007
4. Stephanie Bodoff et al: The J2EE Tutorial, 2nd Edition, Pearson Education, 2006